



JECRCTM
UNIVERSITY
BUILD YOUR WORLD

Faculty of Engineering and Technology

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB MANUAL

Advanced Excel



JECRC UNIVERSITY JAIPUR

Lab Objectives/Aims

Upon successful completion of this lab the student will be able to:

1. To enable students to understand and apply essential features of Microsoft Excel.
2. To develop skills in data entry, formatting, charting, and formula usage.
3. To introduce students to advanced Excel functionalities like pivot tables, lookup functions, and data analysis tools.
4. To encourage the use of Excel for solving real-world problems and preparing reports.
5. To build confidence in creating automated tasks using Excel macros and dashboards.



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Module 1 – Excel Basics

Introduction to Excel

- Interface, cells, rows, columns, sheets
- Data entry (text, numbers, dates)
- **Practical:** Create a student list with names, roll numbers, and classes.

Formatting & Editing

- Font styles, colors, borders, number formats
- Merge & Center, Wrap Text
- **Practical:** Format a marksheet with bold headings & borders.

Basic Formulas

- SUM, AVERAGE, MIN, MAX, COUNT
- **Practical:** Calculate total and average marks for students.

Sorting & Filtering

- Sort A–Z, Z–A, Filter by condition
- **Practical:** Sort a sales list by amount and filter top 10 sales.

Module 2 – Core Excel Functions

Text Functions

- CONCATENATE / TEXTJOIN, LEFT, RIGHT, MID, UPPER, LOWER, PROPER, LEN, TRIM
- **Practical:** Create full names from first & last names.

Logical Functions

- IF, AND, OR, Nested IF
- **Practical:** Grade students based on marks.

Date Functions

- TODAY, NOW, YEAR, MONTH, DAY, DATEDIF

- **Practical:** Calculate employee years of service.

Conditional Formatting

- Highlight rules, color scales, icon sets
- **Practical:** Highlight overdue payments in red.

Module 3 – Data Lookup & Organization

Lookup Functions

- VLOOKUP, HLOOKUP, XLOOKUP (if available)
- **Practical:** Get product price from product list.

INDEX & MATCH

- INDEX basics, MATCH function
- Combining INDEX + MATCH
- **Practical:** Search salary by employee name.

Basic Pivot Tables

- Create pivot table, group data, filters
- **Practical:** Show total salary by department.

Charts (Basic)

- Column, Bar, Pie, Line charts
- **Practical:** Create a monthly sales chart.

Module 4 – Advanced Analysis

Advanced Pivot Tables

- Calculated fields, slicers, drill down
- **Practical:** Compare sales by year and region.

Advanced Charts

- Combo chart, Doughnut, Waterfall, Scatter
- **Practical:** Sales vs. Target combo chart.

Data Validation

- Drop-down lists, custom rules
- **Practical:** Create attendance sheet with Present/Absent list.

What-If Analysis

- Goal Seek, Scenario Manager, Data Tables
 - **Practical:** Loan EMI calculation.
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Module 5 – Data Cleaning & Automation

Data Cleaning Tools

- Remove duplicates, Text to Columns, Find & Replace
- **Practical:** Clean customer contact data.

Power Query Basics

- Import data, transform, merge queries
- **Practical:** Combine sales from multiple sheets.

Basic Macros

- Record macros, assign to button
- **Practical:** Automate report formatting.

Basic VBA Introduction

- Writing simple VBA scripts
 - **Practical:** Highlight salaries above ₹10,000.
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Final Projects & Dashboards

Dashboard Basics

- Planning, layout, linking charts and tables
- **Practical:** Create small sales dashboard.

Interactive Dashboards

- Slicers, timelines, dynamic charts
- **Practical:** Interactive HR dashboard.

Project 1: Business Report

- Combine formulas, charts, pivot tables

- **Practical:** Annual sales & profit report.

Project 2: Personal Finance Tracker

- Expense tracker with visual summary
- **Practical:** Monthly budget vs. actual spend dashboard.

Excel Practical Solutions

Module	Practical Exercise	Exercise Details
Module 1. Excel Basics	Student List	Steps: Enter headings (Name, Roll No., Class) → Fill rows with sample data. Solution: Simple table with 10 rows. Expected Output: Well-aligned student table.
	Marksheet Formatting	Steps: Select headings → Bold → Fill color (light yellow) → Borders for all cells. Expected Output: A neatly formatted table with readable headings.
	Total & Average Marks	Formula: makefile CopyEdit =SUM(B2:D2) → Total Marks =AVERAGE(B2:D2) → Average Marks Expected Output: Each student's total and average calculated automatically.
	Sort & Filter Sales List	Steps: Sort sales column Z→A → Data → Filter → Number Filters → Greater than 500. Expected Output: Only top-selling products appear.
Module 2 – Core Excel Functions	Full Name Creation	Formula: =PROPER(CONCATENATE(A2," ",B2)) Expected Output: "rohan sharma" → "Rohan Sharma"
	Pass/Fail Grading	Formula: =IF(B2>=40,"Pass","Fail") Expected Output: Each student labeled Pass or Fail.
	Employee Service Years	Formula: =DATEDIF(B2,TODAY(),"Y")

		Expected Output: "10" if employee joined 10 years ago.
	Highlight Overdue Payments	Steps: Conditional Formatting → New Rule → =C2<TODAY() → Fill color red. Expected Output: Overdue dates highlighted
Module 3	Data Lookup & Organization	Product Price Lookup Formula: =VLOOKUP(E2,A2:C10,3,FALSE) Expected Output: Correct product price shown.
	Salary Search with INDEX+MATCH	Formula: =INDEX(C2:C10,MATCH(E2,A2:A10,0)) Expected Output: Returns correct salary for given employee name.
	Dept-wise Salary Pivot Table	Steps: Dept → Rows, Salary → Values (Sum). Expected Output: Table showing total salary per department.
	Monthly Sales Chart	Steps: Select Month & Sales columns → Insert → Column Chart. Expected Output: Colorful bar chart showing monthly trend.
Module 4 –	Advanced Analysis	Sales by Year & Region Steps: Year → Columns, Region → Rows, Sales → Values (Sum). Add slicers for filtering. Expected Output: Pivot table showing regional sales per year
	Sales vs. Target Combo Chart	Steps: Insert → Combo Chart → Sales as columns, Target as line. Expected Output: Dual chart comparing actual sales and targets
	Attendance Drop-Down	Steps: Data Validation → List: Present, Absent. Expected Output: Click cell to select Present/Absent.